

The Mechanism of Mass Generating Gravity: From Wheeler's "Mass without Mass" to the Geometric Intuition of QCD and the Higgs Mechanism.

Multiplicative-Entropy-Driven Invariant-Topology Two-Layer Chiral-Space (MITC)

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Funding Declaration: No funding was received

This article is primarily conceptual and discursive in nature. Its aim is not to present a fully formalized mathematical theory, but to articulate a self-consistent physical picture that bridges geometric intuition with quantum field dynamics. We view this picture as a starting point for dialogue and a potential blueprint, and we hope it will attract the attention of researchers in quantum gravity and field theory who are equipped to develop the formal mathematical structure it requires.

1. Abstract

This paper presents the conceptual process of a physical model based on a dual-layer, topologically invariant spatial structure. Starting from a stance of strong physical realism and reductionism, the model views space as a dynamic network of a two-layer foundational space consisting of a sub-Planckian elastic substratum and Planck-scale Spatial Elementary Quanta (SEQs). Building upon Wheeler's view that mass, like gravity, is also an aspect of spatial geometry, the model posits that mass is a spatially compressed state locked by the Higgs mechanism, with QCD serving as its internal dynamics. The Higgs mechanism locks this compressed state, while gravity is the resulting global spatial tensile state induced by the compression. This provides an intuitive explanation for how mass generates gravity, as well as phenomena such as Newton's inverse-square law, the non-shieldability of gravity, the energy scale disparity between QCD and gravity, the nature of dark matter.

2. Introduction

The relationship between gravity and mass is a core issue in gravitational models. General Relativity [1] describes gravity as spacetime curvature, while Quantum Field Theory explains the origin of mass for both elementary and composite particles [2][3][4][5]. However, a fundamental problem remains unresolved between these two theoretical frameworks: Through what specific mechanism does mass generate gravity? That is, the physical process by which the "mass" property of matter is transformed into the "curving" effect of spacetime is still not clear. Building on General Relativity last century, Wheeler proposed "mass without mass" [6], suggesting that matter itself is also spatial geometry. This physical picture is extremely compelling. In the decades since, a diverse range of research programs have developed, exploring avenues such as solitons, elastic spacetime, analogue gravity, loop quantum gravity, causal sets, spacetime thermodynamics, and computational universe models [7]-[48]. (i) Yet, a major obstacle to realizing this vision is how to reconcile this geometric picture with the high-energy dynamic degrees of freedom, such as quantum spin, in the microscopic, high-energy world. (ii) Furthermore, if matter is a geometrically compressed state of space, how is this compressed state maintained to prevent tensor-induced recoil, and how is it compatible with the series of mathematically rigorous, phenomenologically verified models of the Standard Model of Quantum Field Theory? Regarding matter waves and the de Broglie-Bohm theory within existing quantum theory, de Broglie, Bohm, Holland [49][50][51][52][53], and others have conducted significant work.

This model addresses the issue by constructing a hierarchical spatial structure. It uses a sub-Planckian elastic constituent as the underlying substrate. The Space Elementary Quanta (SEQ) serve as topologically stable, quantized network nodes composed of these sub-Planckian constituents. The dynamic motions of quanta, including spin-oscillations, are essentially waves excited on the sub-Planckian constituents. This approach achieves compatibility between quantum spin degrees of freedom and the fiber-like geometric degrees of freedom of space, thereby addressing the first question mentioned above. Subsequently, the model proposes that the Higgs mechanism is essentially a chiral locking mechanism, analogous to a one-way chiral screw mechanism, synthesized at high-energy states from the spacetime metric's Kerr spinor [54] and the fixed-chirality ground-state spin of the SEQ (which, within this model, corresponds to the twistors of Einstein-Cartan Theory (ECT) [55]). This mechanism locks the local spatially compressed state to form mass, providing an intuitive geometric picture for the Higgs mechanism on a realist physical level and aligning with Wheeler's attractive idea of "matter as geometry." The model also offers a geometric intuition for the SU(3) symmetry of QCD color dynamics: it represents the dynamic spherical symmetry that maintains the local compressed state, corresponding to the spherically divergent gravitational field induced by material mass. This

achieves a geometric visualization of the mass-gravity duality.

3. Established Gravitational Phenomena

- 3.1. Newtonian Gravity and the Inverse-Square Law of Action Distance
 - 3.2. Gravity Cannot Be Shielded
 - 3.3. Gravity is Long-Range
 - 3.4. The Strength of Gravity is Far Weaker than that of QCD
 - 3.5. Gravity Generated by Baryonic Mass is Spherically Divergent
 - 3.6. Dark Matter Can Also Generate Gravity
 - 3.7. Gravity Can Warp Space – Light Bending Around Black Holes
 - 3.8. Gravitational Redshift Phenomenon and Gravitational Time Dilation
Does the energy lost from the decrease in photon frequency truly disappear?
 - 3.9. Gravitational Wave Phenomena
 - 3.10. How Do Material Mass or Dark Matter Generate Gravity?
- A theory of gravity must first provide explanations consistent with the above phenomena.

4. Existing Theories of Gravity and Mass

- General Relativity: Gravity and Geometry (hinting at the elastic-like properties of the gravitational field)
- Analogue Gravity
- Modified Gravity
- Loop Quantum Gravity
- Entropic Gravity
- Quantum Field Theory delves into the Higgs mechanism generating mass, with QCD providing the high-energy component. The Standard Model partially explains the origin of mass and is complementary to gravitational theories.

5. Major Problems in Quantum Gravity Research

- First Problem: Incompatibility between geometric degrees of freedom and gauge degrees of freedom. The geometric degrees of freedom in General Relativity are based on 4D spacetime diffeomorphism invariance. In contrast, the Standard Model of Quantum Field Theory, which explains electromagnetic, weak, and strong interactions (QCD) and the Higgs mechanism, relies on degrees of freedom such as spin. The gauge group degrees of freedom could break the 4D differential topological homeomorphism of GR at the Planck scale. This contradiction is the primary obstacle to a unified field theory at the ontological level. Previous research has attempted to introduce dynamic reconnection of geometric degrees of freedom. Theories such as string theory try to bypass ontological realism and provide solutions directly from the perspective of mathematical construction.
- How does the Higgs mechanism generate mass, and how does mass in turn

- generate gravity?
- Why is there such a large disparity in energy scales between gravity and QCD?
- Why is gravity impossible to shield?
- What is the nature of mass and gravity, and through what are they mediated?
- Do quantum fields and the gravitational field share a common substrate?
- Why is it difficult to detect the quantization of energy transmission in the gravitational field?

6. Derivation Process and Ideas of My Proposed Spatial Model, Addressing Existing Problems

6.1. My Personal Preference for Strong Realism Aligns with Physicists Such as Einstein, Wheeler, Schrödinger, and de Broglie.

6.2. Mass and Gravity Require a Common Substrate: What is the Topological Configuration of Space?

A two-layer, 3D topologically invariant space with fixed chirality.

6.2.1 The Necessity of Introducing a Sub-Planckian Layer.

The geometric deformation degrees of freedom of General Relativity and the spin and other degrees of freedom of QFT are difficult to reconcile and coexist at the level of physical reality. Since the quantization of energy at the quantum level occurs at the Planck scale, the underlying substrate mediating gravity is more likely to be smaller than the Planck scale. This construction can actually explain why it is difficult to detect quantization phenomena in gravitational interactions. Understanding the dynamical degrees of freedom of quantum spin as topologically stable collective excitation waves on the sub-Planckian geometric deformation substrate provides an economical scheme for realizing the coexistence of geometric deformation and quantum spin degrees of freedom.

Thus, the configuration of the two-layer foundational space becomes clear: space is composed of an elastic substratum at the sub-Planckian layer (there is no need here to discuss whether this elastic substratum is discretely quantized or continuous, as the quantum phenomena we currently observe are all at the Planck level), while the quantized layer of QFT consists of topologically stable collective excitation waves formed by the sub-Planckian elastic layer.

Similarly, based on the phenomenon that gravity cannot be shielded, the phase operations of the gauge quantum layer cannot shield the gravitational field. Therefore, understanding the gravitational field as composed of components more fundamental than the Planck scale of the quantum fields, the sub-Planckian scale, is a reasonable postulate.

Postulating the existence of sub-Planckian constituents is, at present, one of the simplest and most phenomenologically consistent choices for integrating the degrees of freedom of General Relativity and Quantum Field Theory.

6.2.2 Why Global Topological Invariance is Needed.

Both General Relativity and some analogue gravity theories describe gravity through spatial deformation. Therefore, a fundamental setting is whether space is topologically invariant or topologically variable. The current mainstream, General Relativity itself, in my understanding, within the setting of differential topological invariance of a 4-dimensional manifold and a continuous manifold, allows for constructions like singularities and different 3+1 decompositions. This is essentially an ambiguous zone between 4-dimensional spacetime differential topological invariance and 3-dimensional spatial topological invariance or variability. Many cutting-edge quantum gravity theories tend to view spacetime as a globally topologically variable entity. While this may seem to give the theory greater freedom, I believe this path warrants further discussion from a physical perspective. From an observational standpoint, there is no evidence for 3D definitive spatial topological variability. Considering the constraint of the First Law of Thermodynamics, it is difficult to construct global energy conservation within a topologically variable framework. In my understanding, it is precisely the lack of the strong constraint of 3D global topological invariance in General Relativity that prevents the derivation of the fundamental principle of global energy conservation (the First Law of Thermodynamics). Therefore, setting global topological invariance is a reasonable hypothesis, consistent both with observations and with fundamental thermodynamic laws. Furthermore, if global topological variability were set, hidden problems would actually exist in the current QFT computational framework, such as the issue of unpredictable evolutionary paths. Therefore, setting global topological invariance is more consistent with existing theory. Hence, for the sake of simplicity with respect to phenomena and theoretical self-consistency, I choose to adopt "the 3D topological homeomorphic structure of space remains unchanged during physical evolution" as a fundamental postulate of the model. *(We remark that 4D diffeomorphism invariance allows for time-varying spatial topology; our postulate of global 3D topological invariance is a stronger condition that selects spacetimes with a fixed spatial arena, necessary for unambiguous global conservation laws.)*

At this point, we can roughly conceive of a 3D topologically invariant, quantized configuration of the two-layer foundational space. The Planck-scale Spatial Elementary Quanta (SEQ) forming the quantum network and the spatial substrate of the sub-Planckian elastic layer both remain topologically invariant. The Planck-scale quanta themselves are composed of sub-Planckian constituents, and their dynamical processes are essentially collective excitation waves on these constituents that preserve quantum topological stability.

Matter consists of stable topological structures excited within the quantum network under the influence of QCD, the Higgs mechanism, etc., and are wave packets that resist dispersion effects.

6.2.3 Why Set Space to Have Fixed Chirality.

“Experiments such as those of Prof. Chien- Shiung Wu demonstrated that the weak interaction violates parity symmetry. When this empirical fact is considered together with other chiral asymmetries (e.g., in particle-antiparticle stability and Higgs symmetry breaking), it becomes plausible to hypothesize that space may have an intrinsic chiral structure, which could be the common origin of these phenomena. So how is this chiral characteristic specifically manifested in space? Here, we postulate that the Spatial Elementary Quanta (SEQ) in this model possess a fixed-chirality spin. Under this model, since electrons are experimentally measured to have mass and a spherically divergent gravitational field, it is inferred that electrons are more likely to have a substructure. The observed chiral preferences in related experiments are actually due to different coupling interactions between the electron's structural spin and the fixed-chirality spin of the SEQ. This can also explain the asymmetry in particle-antiparticle stability. From this, a falsifiable prediction is proposed: the magnetic moments of electrons and positrons should have measurable differences because the coupling strength and coupling modes between their respective structural spins and the fixed-chirality spin of the SEQ are different. The role of the SEQ's fixed-chirality spin in the Higgs mechanism will be mentioned in later sections.

6.3. The Energy of the Higgs Field Cannot Explain the Mass-Energy Equation; The Energy Source of Mass Must Largely Come from QCD.

6.4. Analyzing the Geometric Intuition of SU(3) Symmetry

General Relativity, Wheeler's "mass as geometry," and some analogue gravity theories all describe gravity through spatial deformation. If we understand the gravitational field as a tensile state of space, then how does mass generate gravity? An intuitive picture is that mass itself is a compressed state of space. When part of an elastic body is highly compressed, it induces tension in the remaining parts. This physical image is very clear. But why would such a spatially compressed state as mass remain stable and not recoil? And why is the gravity generated by mass spherically divergent?

Because gravity is spherically divergent and mass induces gravity, mass itself must also be a quasi-dynamic spherical structure. Since QCD governs the internal dynamical evolution of massive bodies, the SU(3) symmetry should correspond to dynamic spherical symmetry, where complex numbers represent the compression phase.

According to QFT research, the internal dynamics of baryons are primarily governed by QCD, and the essence of mass-energy is the potential energy of the compressed state. Therefore, mapping QCD to this compression potential energy is a natural deduction. Here, although it is just an assumption, the image is clear. The essence of color interaction should be to characterize this

dynamic equilibrium of the compressed state. The $SU(3)$ symmetry should correspond to the maintenance of dynamic spherical symmetry, making the situation clearer. Analyzing the eight generators of $SU(3)$: geometrically, the two diagonal generators can correspond to global deformation modulation preserving energy volume across multiple dimensions. The remaining six generators can be understood from the perspective of transformation transfers among three orthogonal axes/three color representations and compression phase modulation (3×2). These eight generators ensure that QCD dynamics dynamically maintain quasi-spherical symmetry. This interpretation, though speculative, suggests how the full set of generators could dynamically sustain a quasi-spherical, compressed state. However, there remains one problem: how is the compressed state prevented from recoiling? This is the key step to realizing Wheeler's vision of "mass as spatial geometry."

6.5. Analyzing the Geometric Intuition of the Higgs Mechanism: Kerr Spinor, ECT Twistor, Higgs Mechanism

QFT research has clearly established the Higgs mechanism as the key mechanism for mass generation. Considering the property of spontaneous chiral symmetry breaking in the Higgs mechanism, it is likely the crucial mechanism preventing the release of compression potential energy. If we understand QCD as a spherically dynamic, rotating, and dynamically balanced elastic array – the gluon field – then the Higgs mechanism is the key to preventing its recoil. The Higgs mechanism also has a chiral preference, making the picture almost complete. We use the analogy of screwing on a bottle cap to illustrate the Higgs mechanism: If we take the Kerr spinor as the macroscopic spinor of space, corresponding to the direction of twisting the cap; and the ECT twistor (corresponding to the fixed-chirality spin of the SEQ in this model) as the internal thread of the cap, then twisting the cap in a specific direction can tighten it, thereby locking the compression potential energy corresponding to mass – the compressed spring inside the bottle. Twisting in the opposite direction does not tighten it, corresponding to the instability of antiparticles. In other words, only when the chirality of the spatial spinor and the fixed-chirality spin of the SEQ satisfy a specific matching condition does their coupling produce a non-trivial topological binding potential. This potential unidirectionally and stably "locks" the local spatial compression configuration, manifesting as the rest mass we observe. Conversely, for chirality-mismatched configurations (like antiparticles), this locking potential cannot form effectively, leading to system instability, explaining particle-antiparticle asymmetry. The coupling of the Kerr spatial spinor and the ECT twistor is the Higgs mechanism. The Higgs mechanism and QCD collaboratively generate mass. The Kerr spinor itself also warps space, inducing surrounding spatial tension, which is also nearly spherically divergent but not purely radial. Therefore, even leptons without QCD interactions can acquire mass purely through the Higgs mechanism, but it is

far lower than mass involving QCD. This is because the Higgs mechanism, mediated by the spatial spinor, also warps space and induces geometric compression of its own, albeit not in the nearly radial manner characteristic of QCD compression. The Higgs mechanism is essentially an energy-locking mechanism that emerges from the conditional coupling between a macroscopic spacetime spinor and a microscopic quantum fixed-chirality background.

Note: The above geometric intuitive explanations for QCD and the Higgs mechanism may not be rigorous, but the direction is clear. As long as QCD and the Higgs mechanism are mapped to the geometric intuition of a real space, Wheeler's geometric picture can be made compatible with the QFT Standard Model, achieving unification of quantum field theory and gravitational theory. The critical next step is to formalize these intuitions. I hope this conceptual framework might attract the attention of field theory experts interested in the geometric formulation of fundamental interactions.

6.6. The Nature of Mass and Gravity

At this point, the picture of the nature of mass and gravity is clear. Under the physical picture where space is treated as a topologically invariant elastic body and matter is a topologically protected collective excitation wave packet localized within space itself: Mass is the spatially compressed state locked by the Higgs mechanism, QCD is the dynamic mechanism of its internal compressed state, and the gravitational field is the tensile state of the remaining global space induced by the local spatial compression.

6.7. Why Can't Gravity Be Shielded?

In the above physical picture, gravity is directly mediated by the sub-Planckian elastic substratum. Since the SEQ are also stable topological structures composed of this sub-Planckian elastic substratum, the gravitational field generally cannot detect the resonant frequency signals of the SEQ at the Planck scale, nor can it be shielded through coherent quantum resonance at the Planck scale. This is because the Planck-scale SEQ themselves are composed of the sub-Planckian layer elastic constituents that mediate gravity.

6.8. What is Dark Matter More Likely to Be?

Building on this, combined with the phenomenon that dark matter generates gravitational effects without electromagnetic interactions, it can be inferred that dark matter is more likely a low-compression state of space. It is locked by a low-energy-intensity, Higgs-like mechanism involving a weak coupling between the spatial spinor (Kerr spinor) and the fixed-chirality spin of the SEQ. Its interior lacks structures like the localized structural spins present in ordinary matter mass, which could induce significant local

magnetic field excitation and lead to electromagnetic interactions. This could explain existing phenomena regarding dark matter.

6.9. Physical Intuition for Newton's Inverse-Square Law of Gravity

In this model, the gravitational field generated by matter is mediated by the sub-Planckian layer elastic substratum. The number of gravitational field lines produced by a massive object depends on the number of sub-Planckian elastic constituents on its surface. As gravity diverges in space, the density of field lines decreases with the square of the radius of effect, corresponding directly to Newton's inverse-square law. The image is clear.

6.10. Source and Physical Intuition for the Energy Scale Disparity Between QCD and Gravity

From the above derivation, QCD is a local compressed state of space, with its action confined locally. Gravity is the tensile state of the remaining space induced by the local compression, with its action spread across the entire remaining space. Since the gravitational field is induced by mass, the enormous disparity between the energy scales of QCD and gravity stems from the fundamental difference between a localized compression (QCD) and a global tension (gravity). This explanation is intuitively self-consistent.

6.11. Physical Intuition for Gravitational Redshift and Gravitational Time Dilation

As mentioned earlier, since the SEQ themselves are collective excitation waves – stable topological structures composed of sub-Planckian layer elastic constituents – gravity can propagate directly through the elementary quanta. The dynamical evolution of SEQ generally cannot affect gravitational propagation. However, in reverse, based on the phenomena of gravitational redshift and time dilation, it can be deduced that changes in the gravitational field alter the resonant frequency of the SEQ, because, in this model, the frequency of electromagnetic waves is the resonant frequency of the SEQ. Extrapolating from this phenomenon, one can propose a hypothesis: a mirror-duality relationship exists between metric/gravitational field changes and the resonant frequency of the SEQ. Therefore, the energy lost from the decrease in light frequency is stored in the elastic potential energy of space, achieving energy conservation.

This precisely highlights the advantage of introducing the sub-Planckian elastic layer: if the elastic substrate were on the same level as the fundamental quanta, it would be difficult to explain the physical carrier of energy transfer in redshift. In this model, however, gravity propagates through the underlying elastic medium, while electromagnetism is represented by resonant oscillations of the secondary quantum network. Energy can thus be stored and transformed as elastic potential energy, naturally ensuring energy conservation.

6.12. The sub-Planckian layer in this model can directly correspond to the 3-dimensional space in GR's ADM 3+1 decomposition.

Since computational numerical relativity systems have verified the feasibility of discrete models recovering General Relativity, and because GR's mathematical formalism in most cases does not involve 3-dimensional spatial topological variability, I believe that in these contexts, GR's ADM mathematical form is downward compatible with this model.

7. Discussion and Summary

7.1. Unresolved Questions

What is the specific topological configuration of space? To ensure macroscopic isotropy and Lorentz covariance, is it characterized by random dislocations, multidimensional periodic dislocations, or a packing structure?

7.2. Discussion on Future Development Paths

Although this model offers a clear conceptual picture and remains consistent with observed phenomena, its advancement requires the development of a rigorous mathematical formalism. Furthermore, integrating phenomenological data with Monte Carlo simulations is necessary to gradually obtain structural spectra ranging from leptons to protons and molecules. Even if this direction gains recognition within the community, progress may likely require a considerable amount of time.

8. Statement

The content of this article is a condensed compilation, tailored to meet specific word count and formatting requirements, based on a series of the author's earlier works documenting the thought process. These were previously posted on the preprint.org platform and archived on the zenodo.org manuscript server [56].

This article aims to distill and summarize the core thought process and physical imagery of this research, with a focused emphasis on presenting its unified explanation for the mechanism of "mass generating gravity." Its primary objective is to build upon Wheeler's image of "mass as geometry" by tentatively providing geometric intuitions for QCD and the Higgs mechanism. The author hopes this will attract the interest of field theory experts to develop this direction further. The derivational thought processes of the model's multiplicative entropy theory and time-entropy mapping theory fall outside the scope of this article.

References:

- [1]Einstein, A. (1916). Die Grundlage der allgemeinen Relativitätstheorie. Annalen der Physik, 354(7), 769-822. DOI: <https://doi.org/10.1002/andp.19163540702>
- [2]Gross, D. , & Wilczek, F. . (1984). Ultraviolet behavior of non-abelian gauge theories. Physical Review Letters, 30(26), 1343-1346. DOI: <https://doi.org/10.1103/PhysRevLett.30.1343>
- [3]Politzer, H. D. (1974). Asymptotic freedom: an approach to strong interactions.

- Physics Reports, 14(4), 129-180. DOI: [https://doi.org/10.1016/0370-1573\(74\)90014-3](https://doi.org/10.1016/0370-1573(74)90014-3)
- [4]F. Englert., R. Brout..(1964). Broken Symmetry and the Mass of Gauge Vector Mesons. Physical Review Letters. DOI: <https://doi.org/10.1103/physrevlett.13.321>
- [5]Peter W. Higgs.(1964).Broken Symmetries and the Masses of Gauge Bosons. Physical Review Letters. DOI: <https://doi.org/10.1103/physrevlett.13.508>
- [6]John A. Wheeler. (1957). On the Nature of Quantum Geometrodynamics. Annals of Physics. DOI: [https://doi.org/10.1016/0003-4916\(57\)90050-7](https://doi.org/10.1016/0003-4916(57)90050-7)
- [7]A. Zee, Quantum Field Theory in a Nutshell, Princeton University Press2003, ISBN: 9780691010199
- [8]Ted Jacobson. (1995). Thermodynamics of spacetime: the Einstein equation of state. Physical Review Letters. DOI: <https://doi.org/10.1103/physrevlett.75.1260>
- [9] Gerard't Hooft.(1999)Quantum Gravity as a Dissipative Deterministic System. Classical and Quantum Gravity. DOI:<https://doi.org/10.1088/0264-9381/16/10/316>
- [10] Rovelli, C. (2004). Quantum Gravity.Cambridge University Press.ISBN: 978-0521715966
- [11] Smolin, L. (2001). Three Roads to Quantum Gravity.Basic Books.ISBN: 978-0465078363
- [12] Sorkin, R.D. (2005). Causal Sets: Discrete Gravity. BOOK CHAPTER published in Series of the Centro De Estudios Científicos. Springer, Boston, MA. https://doi.org/10.1007/0-387-24992-3_7
- [13]Witten, E. (1988). Topological Quantum Field Theory. Communications in Mathematical Physics. DOI: <https://doi.org/10.1007/BF01223371>
- [14] Michael S. Turner | Martin White. (1997) CDM Models with a Smooth Component. Physical Review D. DOI: <https://doi.org/10.1103/physrevd.56.r4439>
- [15] Xiao-Gang Wen.(2002).Quantum Orders and Symmetric Spin Liquids.Physical Review B. DOI: <https://doi.org/10.1103/physrevb.65.165113>
- [16] John Archibald Wheeler, Kenneth Ford, John S. Rigden; Geons, Black Holes and Quantum Foam: A Life in Physics. Am. J. Phys. 1 June 2000; 68 (6): 584–585. <https://doi.org/10.1119/1.19497>
- [17]Fotini Markopoulou | Lee Smolin. (2007). Disordered Locality in Loop Quantum Gravity States. Classical and Quantum Gravity. DOI: <https://doi.org/10.1088/0264-9381/24/15/003>
- [18] David Finkelstein. (1969). Space-Time Code.Physical Review, 184(5), 1261-1271.Physical Review. DOI: <https://doi.org/10.1103/PhysRev.184.1261>
- [19]Wayne C. Myrvold. (2020) The Science of $\Theta\Delta cs''$ Foundations of Physics 50,1219–1251. Foundations of Physics. DOI:<https://doi.org/10.1007/s10701-020-00371-3>
- [20]Perez Felipe, S. (2023). Superconducting Field Theory (Theory of Everything). JOURNAL OF ADVANCES IN PHYSICS, 21, 63–72. DOI: <https://doi.org/10.24297/jap.v21i.9464>
- [21]Ali H. Chamseddine | Viatcheslav Mukhanov.(2021) Discrete gravity. Journal of High Energy Physics. DOI: [https://doi.org/10.1007/JHEP11\(2021\)013](https://doi.org/10.1007/JHEP11(2021)013)
- [22]Dmitry Chelkak | Alexander Glazman | Stanislav Smirnov (2016).Discrete stress-energy tensor in the loop $O(n)$ model. DOI: <https://doi.org/10.48550/arXiv.1604.06339>
- [23] X. Fan | T. G. Myers | B. A. D. Sukra | G. Gabrielse,(2023).“Measurement of the Electron Magnetic Moment. Physical Review Letters DOI: <https://doi.org/10.1103/physrevlett.130.071801>
- [24] Manoelito M. de Souza.(2018).Discrete fields, general relativity, other possible implications and experimental evidences. DOI: <https://doi.org/10.48550/arXiv.hep-th/0103218>
- [25]Manton, N., & Sutcliffe, P. (2004). Topological Solitons. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511617034>
- [26]Kalmbach H.E., Gudrun. (2021). Gravity with Color Charges. Journal of Emerging Trends in

- Engineering and Applied Sciences (JETEAS) 11(5):183-189 (ISSN: 2141-7016).
https://www.researchgate.net/publication/348602343_Gravity_with_Color_Charges
- [27]Sim, C. G. (2021). "Gravitational Force Is a Type of Physical Interaction Between Gluon Fields: Molecular Motions of Gases". Physical Science International Journal 25 (6):64-70.
<https://doi.org/10.9734/psij/2021/v25i630266> .
- [28]Wolfram, S. . (2002). A new kind of science. Wolfram Media.
<https://www.wolframscience.com/nks/>
- [29]Raut, U. (2023) A General Relativistic Approach for Non-Perturbative QCD. Journal of High Energy Physics, Gravitation and Cosmology, 9, 917-940. Doi: <https://doi.org/10.4236/jhepgc.2023.94069>
- [30]Ethan Richards. A Complete Unified Theory of Space Compression- Resolving Fundamental Physics Through Mechanical Principles.Academia.edu.
https://www.academia.edu/125340292/_A_Complete_Unified_Theory_of_Space_Compression_Resolving_Fundamental_Physics_Through_Mechanical_Principles
- [31] Ryuya Fukuda, Masataka Iinuma, Yuto Matsumoto, Holger F. Hofmann. (2025). Experimental evidence for the physical delocalization of individual photons in an interferometer.
<https://doi.org/10.48550/arXiv.2505.00336>
- [32] Misner, C.W.; Thorne, K.S. & Wheeler, J.A. (1973). Gravitation, W.H. Freeman, , San Francisco.
- [33] David Cornberg. <https://independent.academia.edu/DavidCornberg>
- [34] John Duffield <https://physicsdetective.com/>
- [35] J. A. Wheeler..(1962) ."Geometrodynamics," Academic Press, Boston,
- [36] Volovik, G. E. . (2009). The Universe in a Helium Droplet. Oxford University Press.
<http://dx.doi.org/10.1093/acprof:oso/9780199564842.001.0001>
- [37]Katanaev, M. O. , & Volovich, I. V. . (1992). Theory of defects in solids and three-dimensional gravity. Annals of Physics, 216(1), 1-28. [https://doi.org/10.1016/0003-4916\(92\)90040-7](https://doi.org/10.1016/0003-4916(92)90040-7)
- [38]Vladimir, Leonov. (2024). V.S. Leonov (1996). Book The Theory of Elastic Quantized Space (EQS).
<http://dx.doi.org/10.13140/RG.2.2.35568.02566>.
- [39] Barceló, C., Liberati, S. & Visser, M. Analogue Gravity.(2005).
<https://doi.org/10.48550/arXiv.gr-qc/0505065>
- [40] HE ZUO-XIU, HUANG TAO.(1974).ON THE COMPOSITE FIELD THEORY AND THE STRATON MODEL (I).. Acta Phys. Sin., 23(4): 40-67. <https://doi.org/10.7498/aps.23.40>
- [41]John M. Baker.(2008) A New Approach to Quantum Gravity from a Model of an Elastic Solid.
<https://doi.org/10.48550/arXiv.0810.4659>
- [42]T.H.R. Skyrme.(1962). A unified field theory of mesons and baryons. Nuclear Physics.
[https://doi.org/10.1016/0029-5582\(62\)90775-7](https://doi.org/10.1016/0029-5582(62)90775-7)
- [43]Shi, J. (2024). Gravity and the Riemannian Hypothesis. Preprints.
<https://doi.org/10.20944/preprints202403.0289.v1>
- [44]J. R. Abo-Shaeer et al..(2001).Observation of Vortex Lattices in Bose-Einstein Condensates.Science292,476-479(2001).DOI:<https://doi.org/10.1126/science.1060182>
- [45]Dunajski, Maciej, Solitons, Instantons, and Twistors, 2nd edn (Oxford, 2024; online edn, Oxford Academic, 20 June 2024), <https://doi.org/10.1093/oso/9780198872535.001.0001>, accessed 18 Dec. 2025.
- [46]Zabusky, Norman & Kruskal, M.. (1965). Interaction of "Solitons" in a Collisionless Plasma and the Recurrence of Initial States. Physical Review Letters. DOI: <https://doi.org/10.1103/PhysRevLett.15.240>.
- [47]Zakharov, V.E. Stability of periodic waves of finite amplitude on the surface of a deep fluid. J Appl Mech Tech Phys 9, 190–194 (1968). <https://doi.org/10.1007/BF00913182>

- [48]Bell, J. S., & Aspect, A. (2004). *Speakable and Unspeakable in Quantum Mechanics: Collected Papers on Quantum Philosophy* (2nd ed.). Cambridge: Cambridge University Press.
- [49]Louis de Broglie.(1924). *Recherches sur la théorie des quanta*, Faculté des Sciences de Paris, Thèse de doctorat soutenue à Paris le 25 novembre 1924. On the theory of quanta, english translation by A.F. Kracklauer. https://fondationlouisdebroglie.org/LDB-oeuvres/De_Broglie_Kracklauer.pdf
- [50]E. Recami, M. Zamboni-Rached, H.E. Hernandez-Figuera, L.A.Ambrosio.(2014).Non-Diffracting Waves: A new introduction. DOI:<https://doi.org/10.48550/arXiv.1408.6503>
- [51]Elbaz C.Sur les programmes d'Albert Einstein et de Louis de Broglie. Une contribution.(2013). ANNALES DE LA FONDATION LOUIS DE BROGLIE.
<https://fondationlouisdebroglie.org/AFLB-381/aflb381m758.htm>
- [52]Bohm, D., & Hiley, B.J. (1993). *The Undivided Universe: An Ontological Interpretation of Quantum Theory* (1st ed.). Routledge. <https://doi.org/10.4324/9780203980385>
- [53]Holland, P. R. (1993). *The Quantum Theory of Motion: An Account of the de Broglie-Bohm Causal Interpretation of Quantum Mechanics*. Cambridge: Cambridge University Press. DOI: <https://doi.org/10.1017/CBO9780511622687>
- [54]Roy P. Kerr.(1963).Gravitational Field of a Spinning Mass as an Example of Algebraically Special Metrics. *Physical Review Letters*. DOI: <https://doi.org/10.1103/physrevlett.11.237>
- [55]Friedrich W. Hehl., Paul von der Heyde., G. David Kerlick., James M. Nester. (1976). General relativity with spin and torsion: Foundations and prospects. *Reviews of Modern Physics*. DOI: <https://doi.org/10.1103/revmodphys.48.393>
- [56]Zou, Z. K. (2025). The Thermodynamic Arrow of Time in a Double-Layer Topology-Invariant Chiral Space with Geometric (GR) and Gauge (QFT) Degrees of Freedom :Time-Entropy Mapping; Mass-Gravity Duality; Metric-Frequency Mirroring. Preprints. <https://doi.org/10.20944/preprints202505.0270.v12>